A Data-Driven Lens into Earthquake Prediction Analysis

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Course: Big Data-CS GY-6513-D

Professor: Dr. Amit Patel

Project Timeline & Milestones

1 **February (1/2) - Earthquake Data Dive**

Uncovered a treasure trove of over 1.2 million historical quake records from Kaggle - time to dive in and uncover insights!

2 **February (2/2) - Laying the Groundwork**

Clearly defined our project goals, scope, and key evaluation metrics to keep us on track and make an impact.

3 **March - Cleaning & Preprocessing**

Loaded the dataset directly into PySpark DataFrames; meticulously cleaned and normalized (filled missing depths/magnitudes, standardized timestamps, removed duplicates), encoded categoricals, and scaled numericals.

4 **April - Modeling & Analysis**

Trained Random Forest, SVM, and Logistic Regression models for earthquake impact analysis; created epicenter scatter plots, hexbin density maps, and state-wise countplots using Matplotlib for exploratory visualization.

5 **May - Evaluating & Visualizing**

Validated our models on a hold-out set and assessed performance metrics; then built an interactive Tableau dashboard to showcase the insights.

Problem Statement

Current Limitations

Earthquake forecasting has long been a challenging problem for scientists and disaster management teams. Current approaches often lack the precision and reliability needed to effectively prepare communities for major seismic events. Despite advancements in seismology and monitoring technologies, predicting the timing, location, and magnitude of earthquakes remains an elusive goal.

- Less than 1% of global earthquakes are predicted with useful accuracy today.

Project Goal

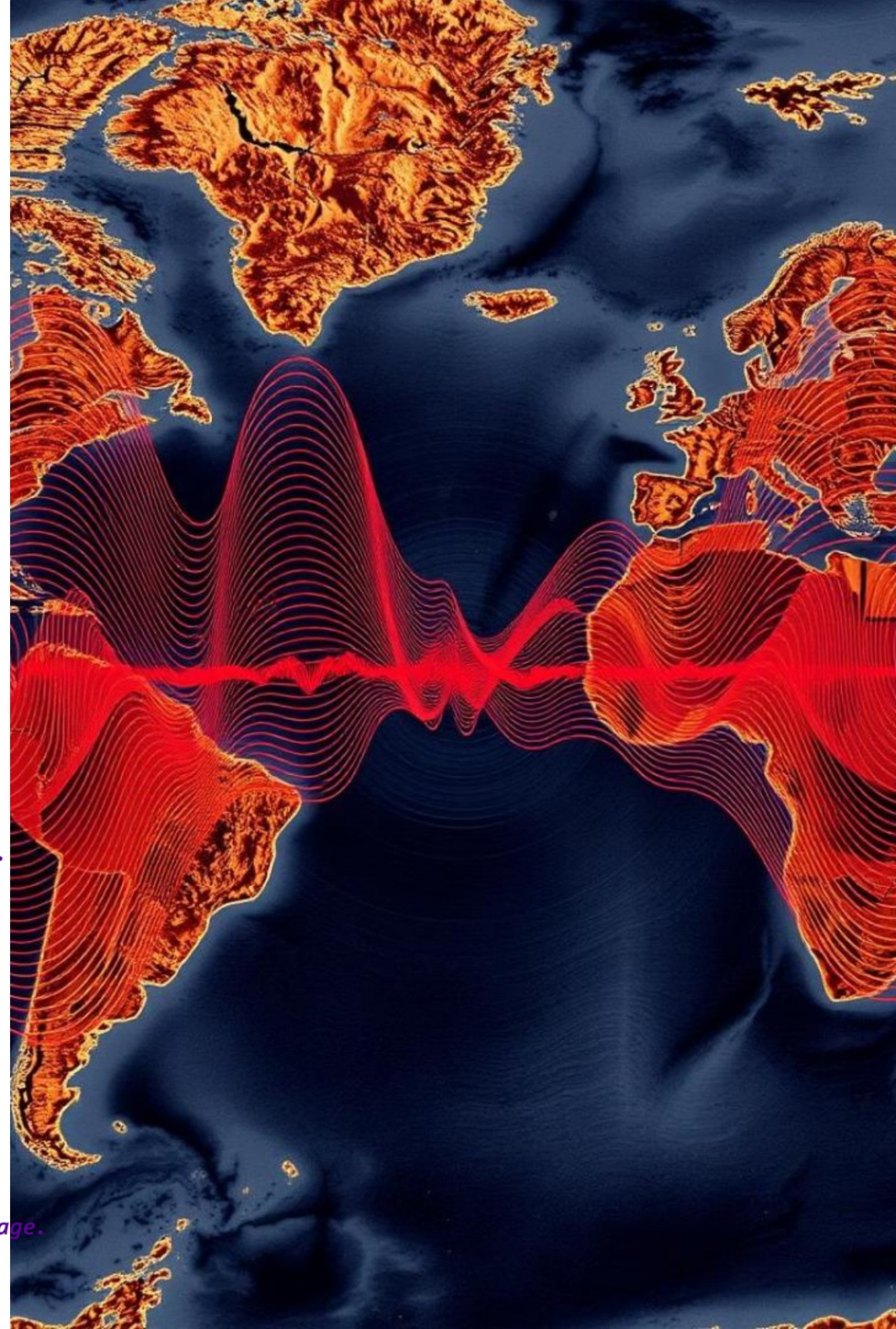
To address these limitations, our team is embarking on an ambitious project to leverage big data and cutting-edge machine learning techniques. By uncovering hidden patterns and relationships within a vast trove of historical earthquake data, we aim to develop more accurate and actionable forecasting models. Our ultimate objective is to equip decision-makers with the insights needed to enhance disaster preparedness and risk mitigation strategies worldwide.

- Utilizing over 30 years of seismic data and 1M+ global earthquake records to train scalable ML models.

Impact

Improving earthquake prediction capabilities could have profound societal benefits. Timely and reliable forecasts would enable emergency responders to mobilize resources more effectively, allow businesses and infrastructure owners to take proactive protective measures, and empower communities to better safeguard lives and property. By delivering these critical insights, our project has the potential to save countless lives and reduce the devastating economic toll of natural disasters.

- Even a 10-minute early warning can reduce earthquake fatalities by up to 50% and save billions in damage.



Project Objectives

Our earthquake prediction project has several key objectives that will guide our work and help us achieve meaningful impact. At the core of our efforts is a commitment to large-scale data analysis, leveraging historical seismic activity data to uncover hidden patterns and relationships.

To handle the immense volume and complexity of this data (1m+ records), we will utilize the power of Apache Spark for efficient processing and analysis. By harnessing the distributed computing capabilities of Spark, we can rapidly extract insights from massive datasets, laying the groundwork for our predictive modeling efforts.

Building on this foundation, we will develop advanced machine learning models specifically tailored for earthquake forecasting. Drawing on the latest advancements in AI, we aim to create predictive algorithms that can anticipate the timing, location, and magnitude of seismic events with greater accuracy than current approaches.

- Targeting a 10–20% improvement in forecast precision over traditional statistical methods.

Finally, we recognize the importance of delivering these insights in a clear and actionable manner. To that end, we will invest significant effort into creating insightful data visualizations and interactive tools that empower decision-makers to make informed choices about disaster preparedness and risk mitigation strategies.

- Designing dashboards that can reduce decision-making time by up to 40% during emergencies.

Dataset Overview

Source & Size

Kaggle dataset with 1 to 1.2 million earthquake records (1990-2023).

# time time in milliseconds	♙ place geographical location	♙ status Represents the current state or condition of the event, which could be reviewed or automatic	# tsunami Relates to a series of large ocean waves typically caused by an underwater disturbance, often associated with	# significance Denotes the importance or impact level of the event, which could be used to assess the potential consequences
631b 1691b	531130 unique values	reviewed 94% automatic 6% Other (15212) 0%	0 1	0 29
631153353990	12 km NNW of Meadow Lakes, Alaska	reviewed	0	96
631153491210	14 km S of Volcano, Hawaii	reviewed	0	31
631154083450	7 km W of Cobb, California	reviewed	0	19
631155512130	11 km E of Mammoth Lakes, California	reviewed	0	15
631155824490	16km N of Fillmore, CA	reviewed	0	134
631155853760	16km N of Fillmore,	reviewed	0	118

Key Features

Date & Time

When did the shaking start? Our models will pinpoint the precise timing of seismic events to the second.

Coordinates

Where on the map did the earthquake hit? We'll zero in on the geographic epicenter to guide emergency response.

Depth & Magnitude

How strong was the quake, and how deep did it originate? These factors are critical for understanding the potential impact.

Location Details

Beyond the numbers, we'll provide rich contextual information about the affected regions and communities.

Technical Challenges

- *Data Quality and Noise*

Historical seismic data often contains inconsistencies and outliers. We applied cleaning and type casting techniques in PySpark to ensure reliable analysis.

- *Handling Missing Values*

Missing or null entries were addressed using median-based imputation and data filtering to ensure robustness.

- *Feature Encoding and Transformation*

Applied `StringIndexer` *and* `OneHotEncoder` *for categorical features; used* `VectorAssembler` *to create ML-ready feature vectors.*

- *High-Dimensional Feature Space*

Selected relevant seismic predictors (e.g., magnitude, depth, location) to reduce redundancy and improve model performance.

- *Model Optimization*

Tuned hyperparameters and validated models using precision, recall, and F1-score to select the best performing classifiers.

System Workflow Overview

Start

Define objectives and secure reliable data source

Data Collection

Gather seismic records from 1990 to 2023, 1.2M entries

Data Preprocessing

Clean, typecast, and impute missing values using PySpark

Exploratory Analysis

Generate spatial and intensity visualizations for insight

Machine Learning

Train and evaluate models using Spark MLlib classifications

Visualization

Create Tableau dashboards with heatmaps and frequency charts

Data Cleaning & Feature Engineering

- *Ingested 30+ years of global earthquake records using* `earthquakes-1990-2023.csv` *with schema inference for robust type detection.*
- *Cast all numeric features (*`latitude`*,* `longitude`*,* `depth`*,* `tsunami`*,* `significance`*,* `magnitudo`*) to* `DoubleType` *and applied distributed median imputation using* `percentile_approx` *to handle missing values at scale.*
- *Removed non-predictive attributes (*`time`*,* `date`*,* `place`*) to reduce feature noise and dimensionality.*
- *Encoded categorical columns (*`state`*,* `data_type`*,* `status`*) using StringIndexer followed by OneHotEncoder, generating sparse binary vectors compatible with Spark ML pipelines.*
- *Assembled numerical and encoded categorical features into a single* `features` *vector using* `VectorAssembler`*, preparing data for efficient model training.*
- *Final dataset persisted as a Spark DataFrame with ~130K rows, ready for cross-validation and classification.*

Scalable Earthquake Classification Using Distributed ML (Random Forest, SVM, Logistic Regression)

- Trained and evaluated three classifiers:
- Random Forest • Support Vector Machine (SVM) • Logistic Regression
- All models used a common feature pipeline with encoded categorical variables and normalized numeric inputs
- Applied manual stratified sampling to ensure balanced representation of rare high-magnitude earthquakes (~2% of data)
- Performed K fold cross-validation (5 times) for each model with grid search over key hyperparameters:

`regParam`, `maxIter` (for SVM & Logistic Regression)

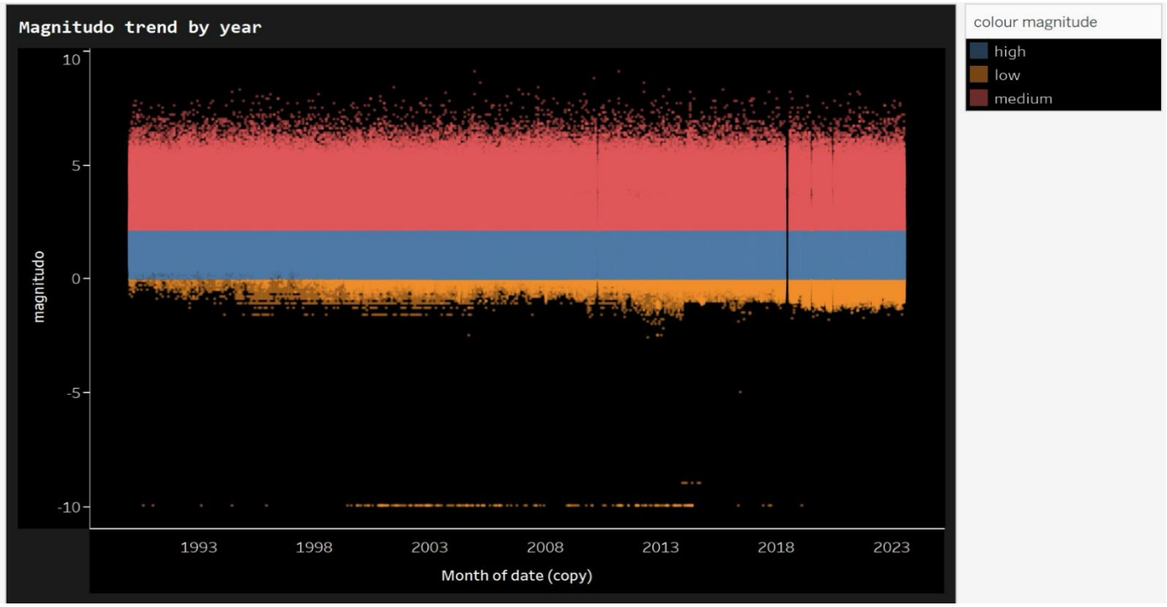
`numTrees`, `maxDepth` (for Random Forest)

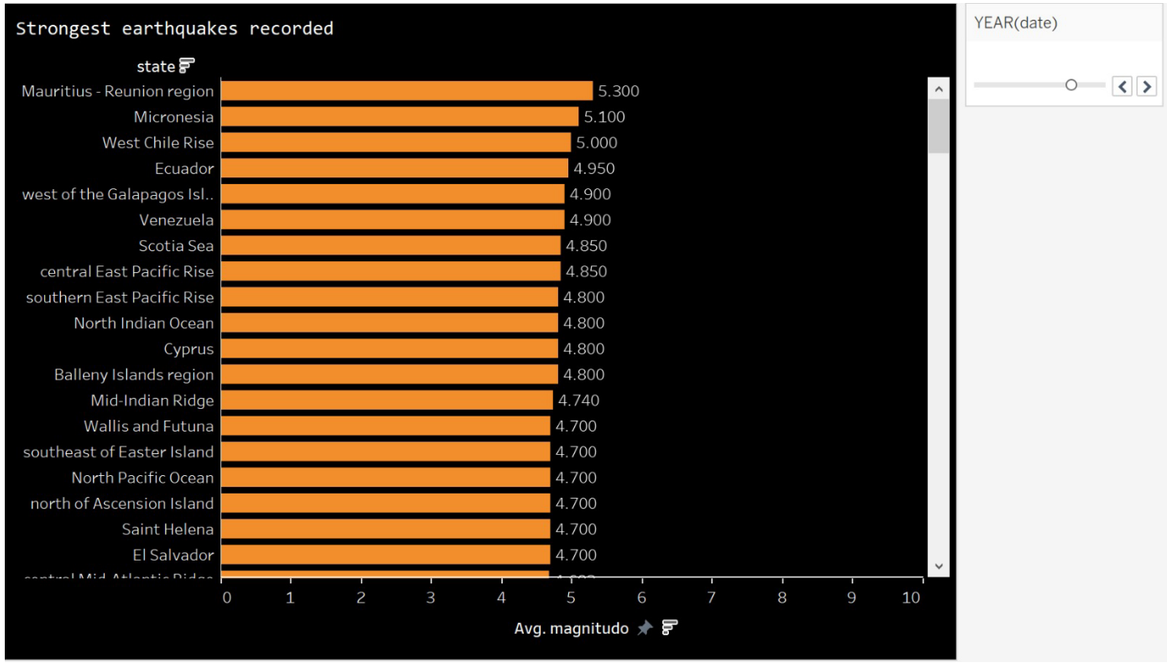
- Compared model performance using:

Accuracy, F1 Score, Precision, Recall, and Confusion Matrix

- SVM performed best based on Accuracy with a score 99.63%

Evaluation & Visualization Insights





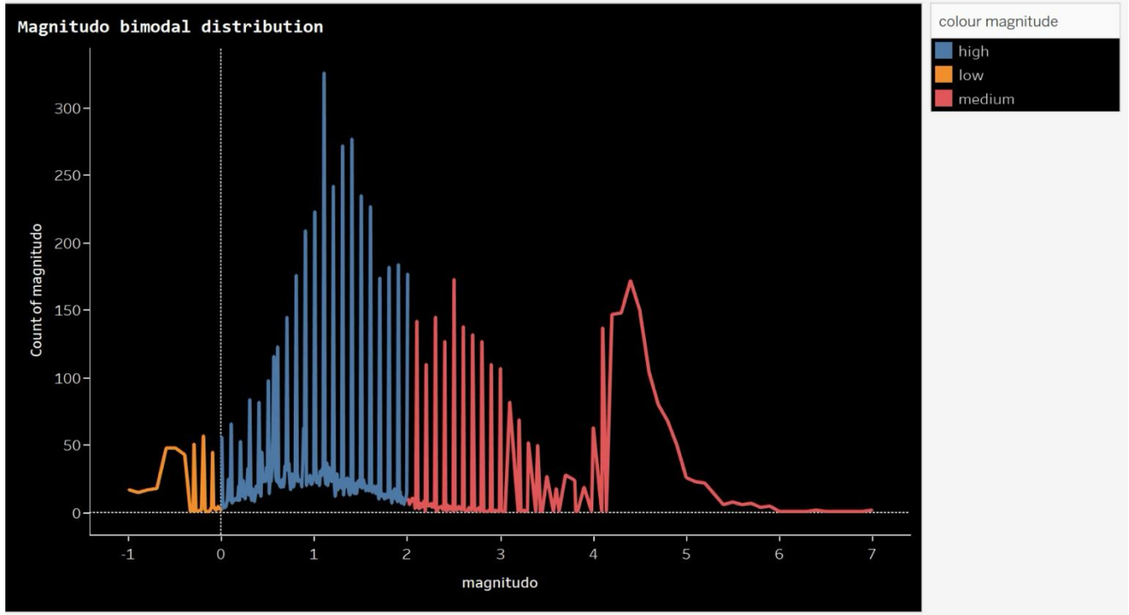
Seismic Activity by Year (1990–2023)

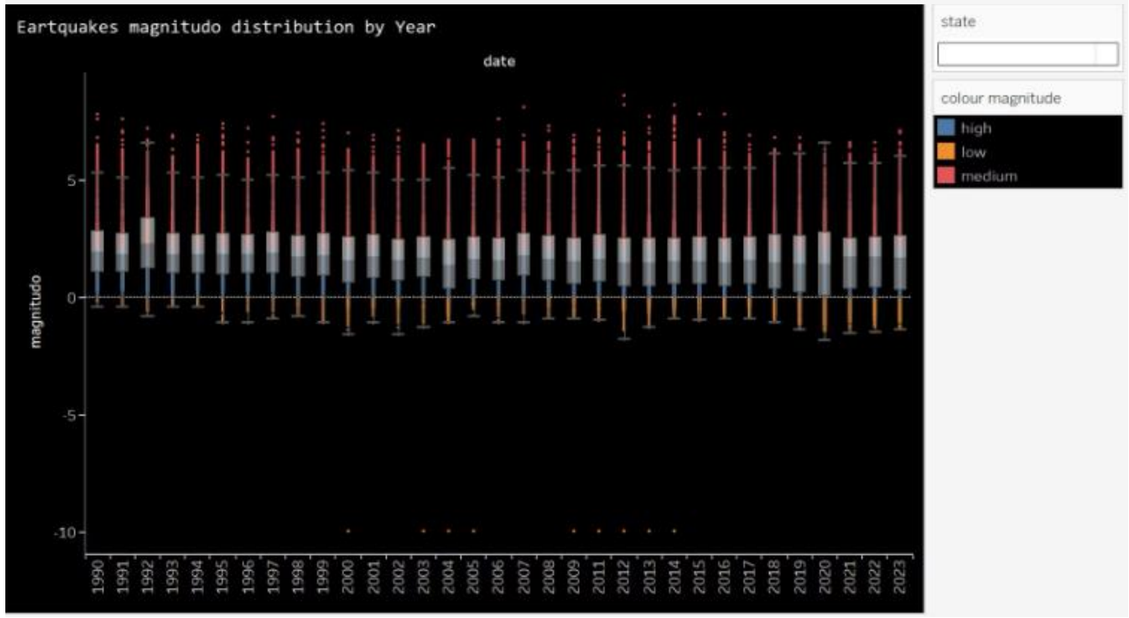
This chart shows earthquake magnitudes over time, color-coded by severity. It helps visualize frequency and trends in high-, medium-, and low-magnitude earthquakes.

Top Regions with Highest average Earthquake Magnitude

Horizontal bar chart ranking regions by average earthquake magnitude. Useful for identifying seismic hotspots with consistently high-impact events.

Evaluation & Visualization Insights





Bimodal Distribution of Earthquake Magnitude

The histogram reveals two peaks in earthquake magnitudes — a classic bimodal pattern.

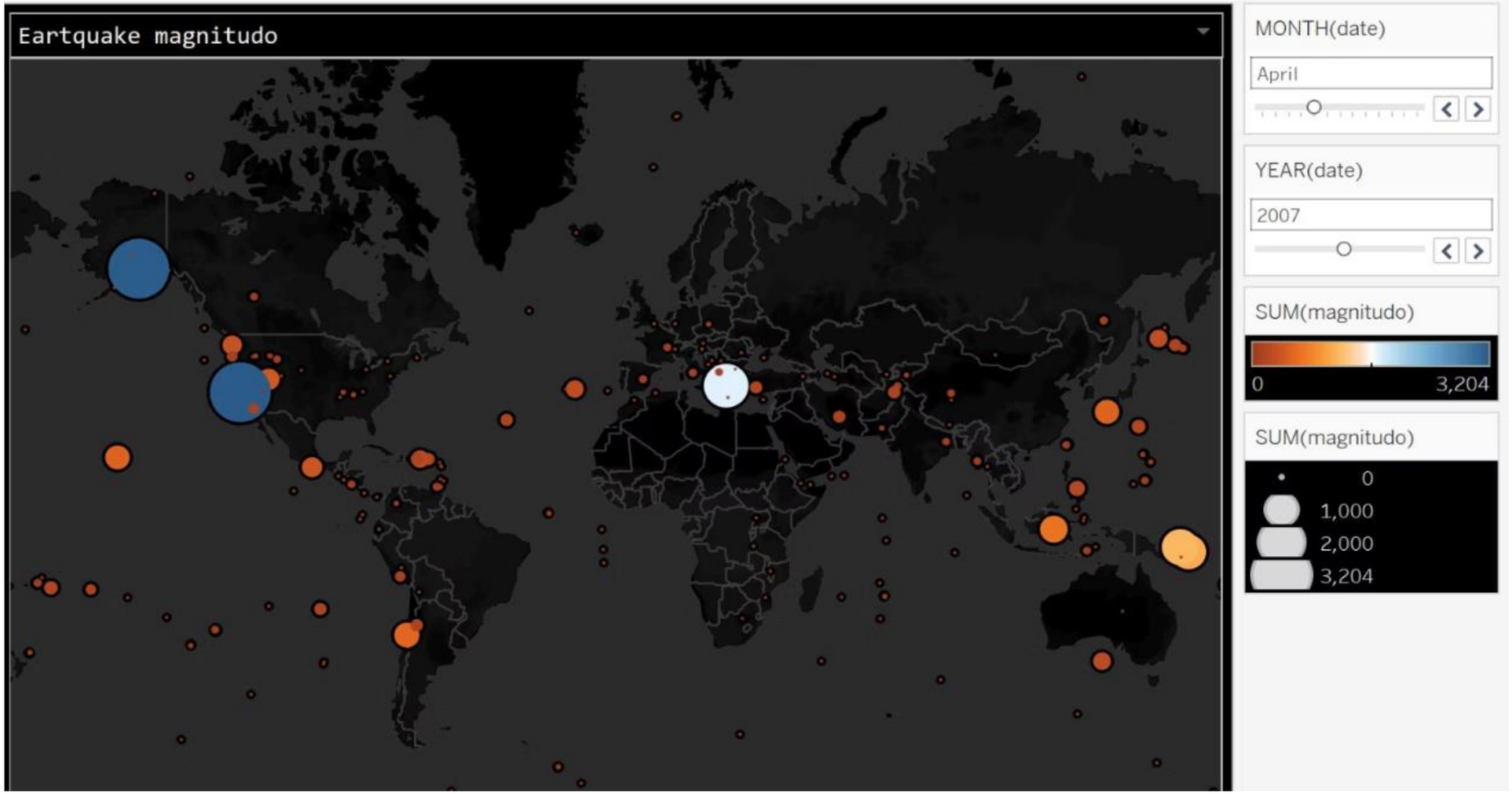
This indicates distinct clusters of weak and strong seismic events in the dataset.

Annual Distribution of Earthquake Magnitudes

Boxplots show median, IQR, and outliers of magnitudes per year.

It highlights yearly variability and the persistence of extreme events.

Evaluation & Visualization Insights



World Map of Earthquake Magnitude (Year & Month Filtered) A geospatial heatmap showing earthquake intensity across regions. Bubble size and color reflect total magnitude per region, filtered by time.

Conclusion & Future Directions

Scalable Pipeline

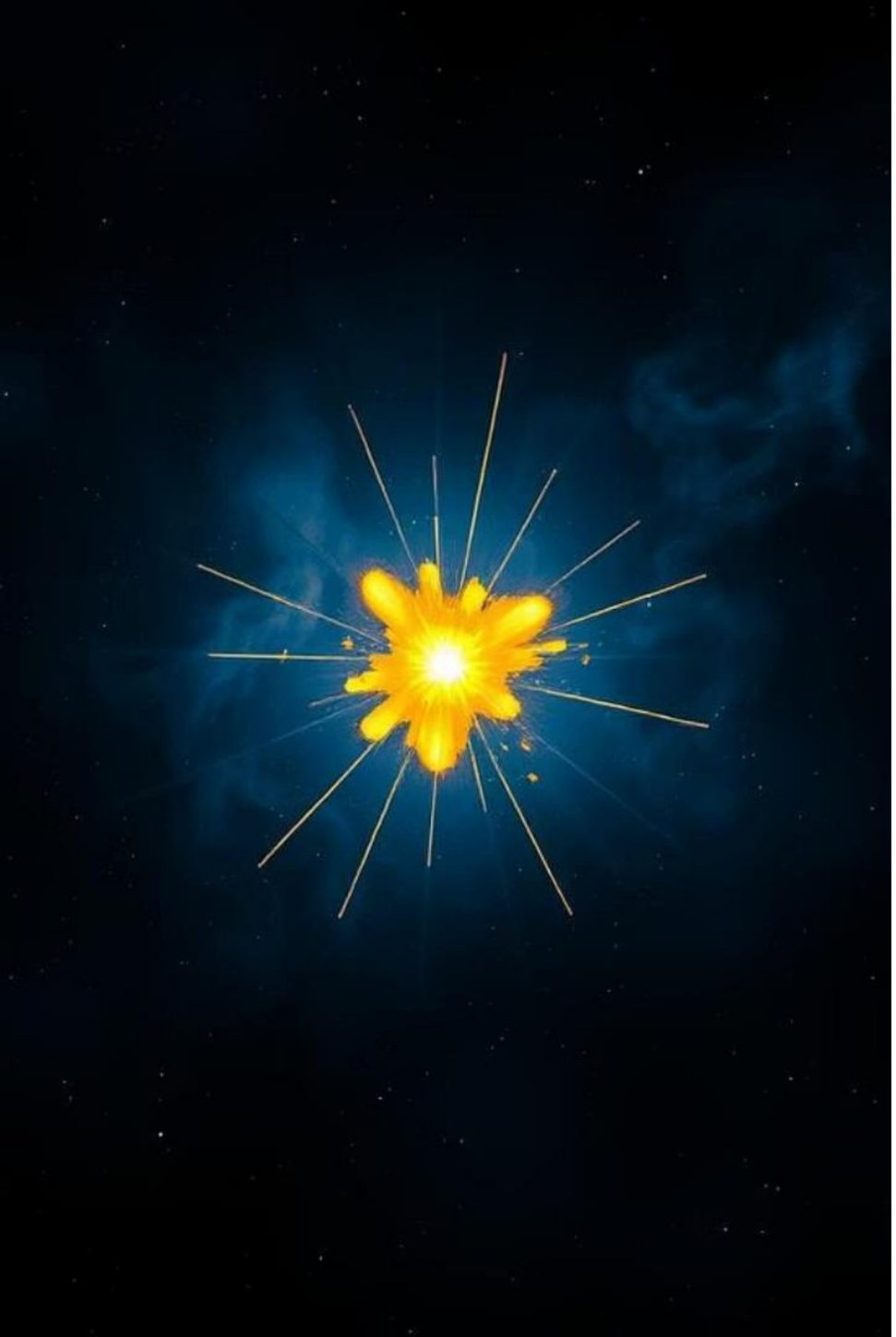
Through this project, we've built a robust, end-to-end data analysis workflow to tackle the challenge of earthquake prediction. By ingesting and processing over 1.2 million seismic records spanning three decades, we've developed a scalable system that can efficiently handle large-scale earthquake data.

Insights Gained

Our deep dive into the earthquake dataset has yielded valuable insights that enhance our understanding of seismic patterns and predictability. We've uncovered hidden trends, identified key features, and gained critical knowledge to power our predictive models.

Future Work

Looking ahead, we plan to further expand the capabilities of our earthquake analysis system. Key priorities include integrating real-time seismic data streams, applying our techniques to other types of natural disasters, and continuously enhancing the robustness and accuracy of our predictive models.



Thank You

We appreciate your time and attention throughout this earthquake prediction analysis journey.

Feel free to ask questions.